C PROGRAMMING FROM SCRATCH · EPISODE 13

ASCII and Escape Sequences

Characters, codes and the backslash

KEY POINTS

- Codes 0–31 and 127 are control characters; 32–126 are printable. 'A' is 65, 'a' is 97, '0' is 48.
- A character and its code are two views of one value: %c prints the character, %d prints the code.
- An escape sequence is a backslash plus a character (or an octal / hex code), and it is always one character.
- Every string ends with a hidden \0; strlen counts the characters before it.

01 The ASCII table

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ASCII defines **128 characters, codes 0 … 127**. Codes **0–31** and **127** (DEL) are control characters: they print no symbol. Code **32** is the space, and **32–126** are printable: letters, digits and punctuation.

CHARACTER	DECIMAL	OCTAL	HEX
`'A'` … `'Z'`	65 … 90	101 … 132	41 … 5A
`'a'` … `'z'`	97 … 122	141 … 172	61 … 7A
`'0'` … `'9'`	48 … 57	60 … 71	30 … 39
newline `'\n'`	10	12	A

KEY IDEA

Lowercase = capital + 32. The character `'0'` is not the number 0: its code is 48.

02 Characters as numbers

▶ VIDEO 02:57

char_codes.c — C SOURCE

```c
#include <stdio.h>

int main()
{
    printf("%c\n", 'a');
    printf("%d\n", 'a');
    printf("%c\n", 97);
    return 0;
}
```

03 Escape sequences

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A backslash changes the meaning of the character after it, and the pair becomes **one character**.

ESCAPE	MEANING	ESCAPE	MEANING
`\n`	newline	`\a`	alert (beep)
`\'`	single quote	`\b`	backspace: cursor back one place
`\"`	double quote	`\r`	carriage return: start of the line
`\\`	one backslash	`\t`	tab: next stop (every 8 columns here)
`\?`	question mark (old trigraphs)	`\v` `\f`	vertical tab, form feed (rare)

cursor.c — C SOURCE

```c
#include <stdio.h>

int main()
{
    printf("abcdef\b\bxyz\n");   // backspace twice, then overwrite
    printf("abcdef\rxy\n");      // back to the start of the line
    printf("a\tb\tc\n");         // tab stops every 8 columns
    printf("ab\tcd\tef\n");
    printf("abc\tdef\tghi\n");
    return 0;
}

/* In the console window this shows:
abcdxyz
xycdef
a       b       c
ab      cd      ef
abc     def     ghi
*/
```

04 Octal and hex escapes

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`\ddd` is 1–3 octal digits, `\x` is followed by hex digits: `'\130'` = 1×64 + 3×8 + 0 = 88 = `'X'`, and `'\x30'` = 3×16 + 0 = 48 = `'0'`.

05 The hidden \0 and strlen

▶ VIDEO 07:35

lengths.c — C SOURCE

```c
#include <stdio.h>
#include <string.h>

int main()
{
    printf("%zu\n", strlen("abcdef"));
    printf("%zu\n", strlen("C:\apps\test\007\new.c"));
    printf("C:\test\test.c\n");
    printf("C:\\test\\test.c\n");
    return 0;
}
```

WATCH OUT

`"C:\apps\test\007\new.c"` shows 22 symbols, but `\a`, `\t`, `\007` and `\n` are one character each, so `strlen` gives 16. Write Windows paths with double backslashes: `"C:\\test\\test.c"`.

CODE 6 files on GitHub

github.com/abitclearer/c-programming-from-scratch/tree/main/13-ascii-and-escape-sequences